

DHARANI KOMMIREDDI

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EDUCATION

MS in Data Science - DePaul University, Chicago, IL
Concentration: Healthcare Analytics · GPA: 3.6

Sept 2024 - June 2026

B.Tech in Computer Science Engineering - Malla Reddy Engineering College, India
Major: Computer Science, Minor: Data Science

Nov 2020 - June 2024

PROFESSIONAL EXPERIENCE

Healthcare Data Analytics Extern

Apr 2026 - Jun 2026

TruBridge (via Extern)

- Analyzed 50,057 hospital admissions and found that medium-term stays (11-30 days) dominate bed capacity while departmental routing creates highly variable resource consumption (Chi-Square $p < 0.001$), identifying early admission signals that could shift bed management from reactive to predictive.
- Cleaned and preprocessed public healthcare data in Python (Google Colab), applying Min-Max scaling and log transformations; built a Power BI dashboard presenting LOS distribution, deposit skew, and departmental volatility to non-technical stakeholders.
- Earned Top 10% Performer recognition; presented capacity-planning recommendations in a structured review with TruBridge analytics professionals.

Marketing & Data Analyst Intern

Jun 2025 - Sep 2025

Gabriel AI | Chicago, IL

- Automated LinkedIn competitive intelligence reporting in Python (Pandas) and Excel as part of a small cross-functional team, cutting ~8 hours/week of manual work and standardizing how the team tracked competitor activity.
- Built Tableau dashboards covering funnel performance, channel attribution, and growth KPIs; presented weekly insights to founders and GTM leads that contributed to a ~20% increase in inbound leads over the quarter.
- Ran ICP analysis in SQL and Google Sheets to segment and score prospects, recommending a shift toward the highest-performing segments to improve demo booking efficiency.

Data Science Intern

Jun 2023 - Dec 2023

Innodatatics | Hyderabad, India

- Built and refactored Python ETL pipelines on client datasets under NDA, cutting end-to-end preprocessing runtime by ~42% through batched transformations and reusable cleaning functions.
- Applied SQL for data extraction and preprocessing (normalization, one-hot encoding, feature engineering) to improve ML model inputs, lifting classification accuracy from baseline by ~34.6%.

SKILLS

Programming & Tools: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, R, Tableau, Power BI, Excel (Pivot Tables, VLOOKUP), Jupyter, Git/GitHub

ML & Analytics: Random Forest, XGBoost, Logistic Regression, Feature Engineering, EDA, SHAP, Fairness Analysis, Model Evaluation (ROC-AUC, F1, RMSE), ETL Pipelines

Healthcare Domain: MIMIC-IV, Readmission Risk, LOS Prediction, SDOH, Patient Risk Stratification, ICD-9/ICD-10 (CCSR), HIPAA-Aligned Data Handling, CDC HI-5 Framework

Certifications: IBM Data Science Professional · Stanford - Intro to Statistics · Coursera - Business Analysis & PM · NASBA - AI & Business Strategy · TruBridge Top 10% Performer

PROJECTS & RESEARCH

30-Day Readmission Prediction with Fairness-Aware Thresholds | *MIMIC-IV, Python, Scikit-learn*

- Built an end-to-end ML pipeline on MIMIC-IV clinical data (546K admissions, 60,946 labeled cohort); engineered 355 features from ICD-9/ICD-10 codes mapped to CCSR categories, demographics, LOS, and discharge disposition with patient-level train-test split to prevent data leakage.
- Benchmarked 12 models; selected Random Forest (AUC 0.6506, 74.2% recall) for minimal overfitting (train-test AUC gap 0.059); SHAP analysis identified prior admission count, LOS, and discharge location as top predictors.
- Conducted fairness audit across race, gender, insurance, and age subgroups; applied stratified decision thresholds per racial group, reducing racial FNR disparity by 86% (48.2pp to 6.6pp) without retraining the model or sacrificing overall AUC.

Cardio Cast: Cardiovascular Risk & Stroke Prediction | *DePaul DSC 540, Python, Scikit-learn, Tableau*

- Built a dual-dataset ML pipeline on UCI heart disease (303 records) and stroke (5,110 records) data; benchmarked 5 models across 5 random states and achieved 94.1% stroke accuracy (AdaBoost) and 0.89 AUC-ROC (XGBoost), with age, hypertension, and glucose emerging as top predictors consistent with clinical literature.

Cloud EMR & Telemedicine Platform | *Published in IRJMETs Vol. 6, IF: 7.868* | [Link](#)

- Designed a HIPAA-aware cloud EMR prototype with three role-based user tiers, blockchain-based interaction logging, and a Bag-of-Words ML classifier for record categorization.